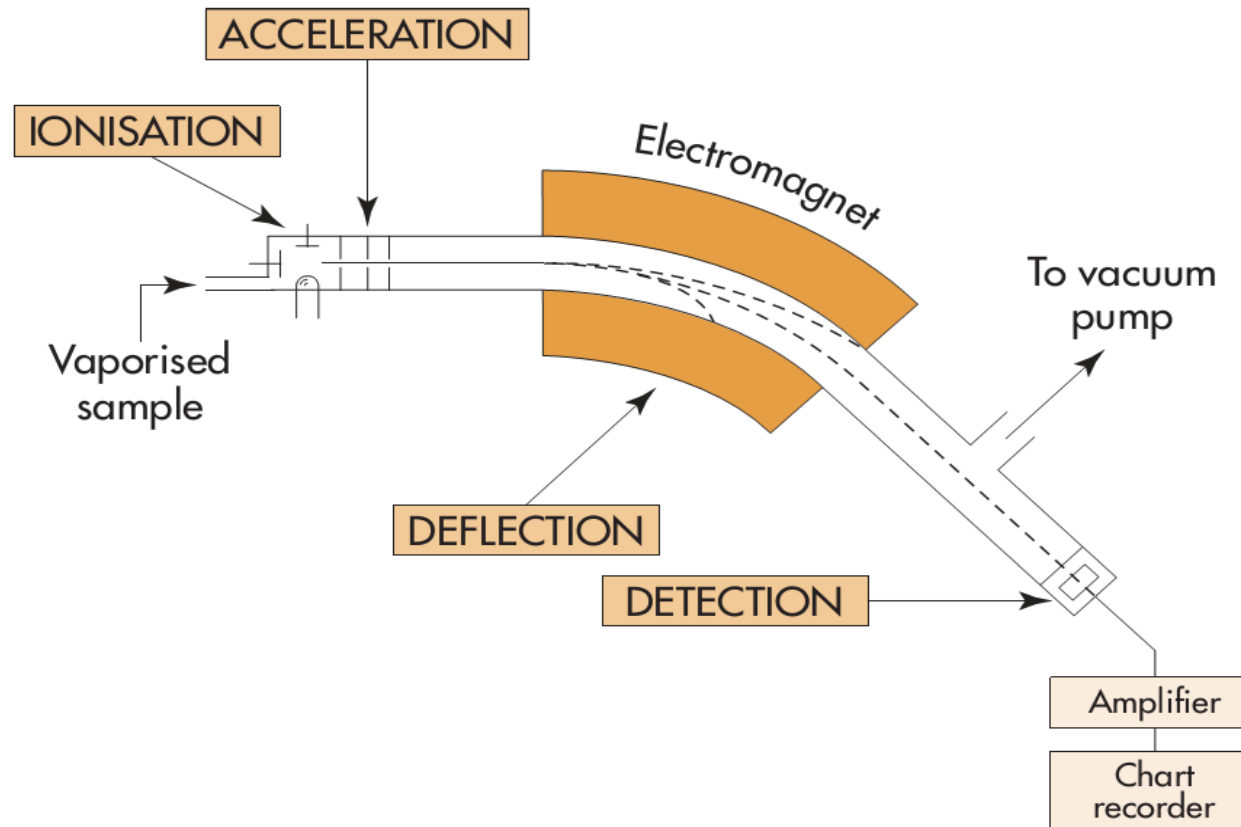


Mass Spectrometry



KEYPOINTS

Principles

- Charged molecules or molecular fragments are generated in a high-vacuum region, or immediately prior to a sample entering a high-vacuum region, using a variety of methods for ion production. The ions are generated in the gas phase so that they can then be manipulated by the application of either electric or magnetic fields to enable the determination of their molecular weights.

Applications

- Mass spectrometry provides a highly specific method for determining or confirming the identity or structure of drugs and raw materials used in their manufacture.
- Mass spectrometry in conjunction with either gas chromatography (GC-MS) or liquid chromatography (LC-MS) provides a method for characterising impurities in drugs and formulation excipients.

Introduction

- ☐ A mass spectrometer is one of the most powerful analytical tool available to scientists today, it is not a spectroscopic technique (i.e. it does not involve the measurement of light radiation).
- ☐ It determines the weight of the molecules by estimating the mass to charge ratio (m/e) of ions.
- ☐ In this technique, molecules are bombarded with electrons; high energized positive ions are produced which are further breakdown to produce small fragments.
- ☐ These ions are separated in magnetic or electric field according to their mass to charge ratio.

Introduction

- In the case of MS, the analytes (drug substances and other organic compounds) are bombarded with electrons in a vacuum inside the mass spectrometer.
- This leads to some of the molecules being ionized either to positive ions (molecules lose an electron) or negative ions (molecules take up an electron), called **molecular ions**.
- The mass of an molecular ion is equivalent to the mass of the original molecule because the weight of an electron that is either lost or taken up is incredibly small.

Introduction

- In some cases, molecular ions are unstable and very quickly split in to smaller fragments as chemical bonds in the molecules are broken.
 - This process is called fragmentation.
- Some of the fragments remain ionized (fragment ions), while others lose their charge

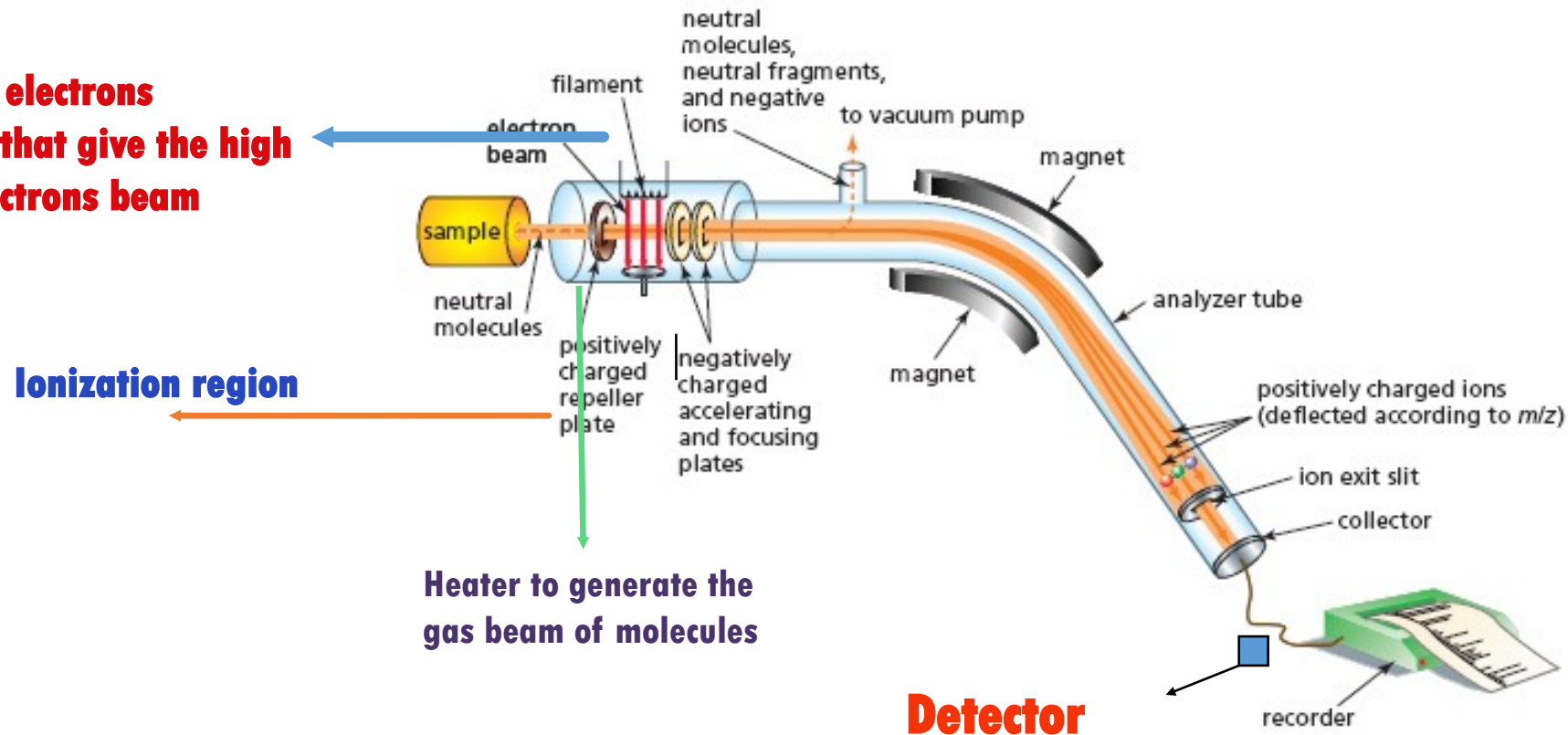
Introduction

- ✚ The appearance of a **mass spectrum**, both in terms of which **ionic species** are observed, as well as **their abundances**, serves as the basis for **compound identification**.
- ✚ The information derived from a mass spectrum is often combined with that from other analytical techniques,
 - ➡ **infrared spectroscopy** and
 - ➡ **nuclear magnetic resonance spectroscopy**, **to generate structural assignments for organic molecules**

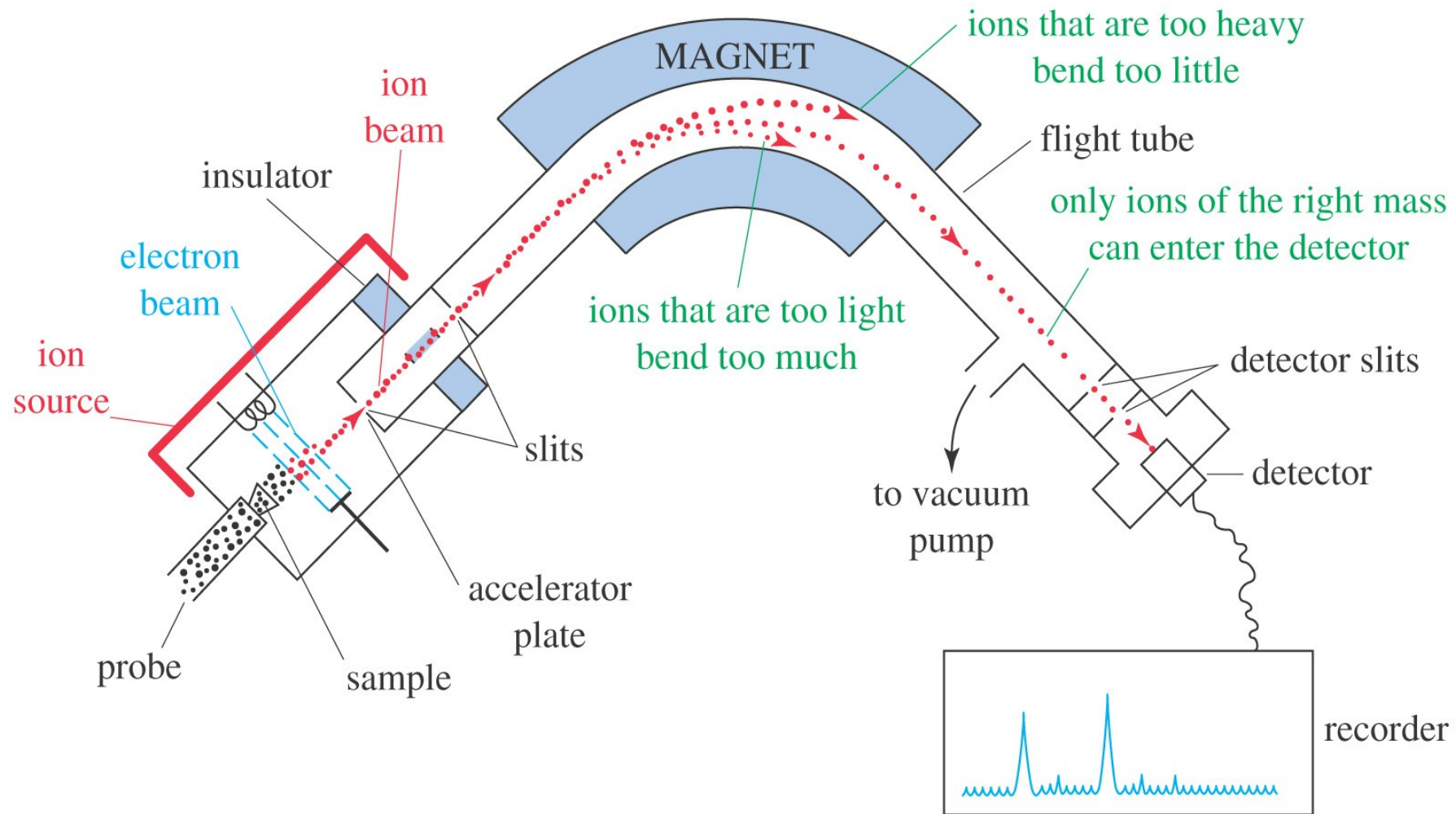
Instrumentation

- The components of mass spectrometer are displayed diagrammatically as below

This is the electrons generator that give the high energy electrons beam



Instrumentation



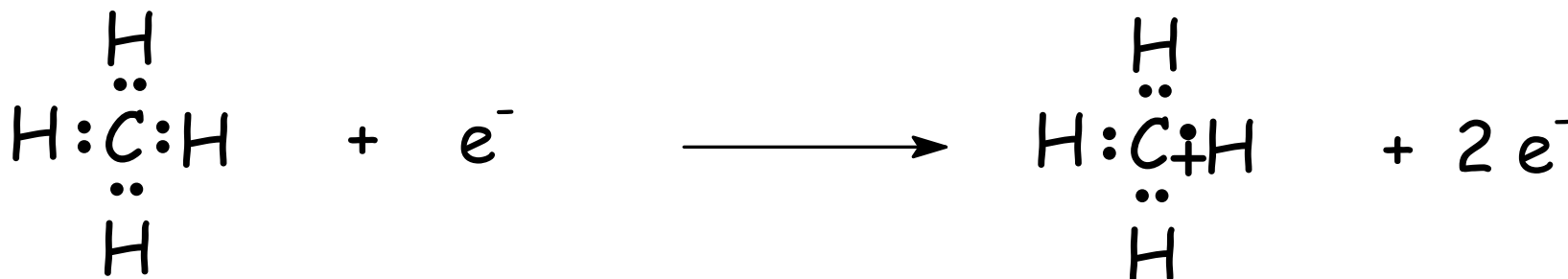
Fundamental Processes Involved in MS

- 1. A sample of the substance to be analysed is injected into the mass spectrometer**
 - **The sample is vaporized**
 - **Volatile substances will vaporize easily in the very low pressure environment used in a mass spectrometer**
 - **Other substances may need to be heated in order to vaporize them**
- 2. A stream of gaseous molecules enters an **ionization chamber** where it is ionized by **different methods**.**

Fundamental Processes Involved in MS.....

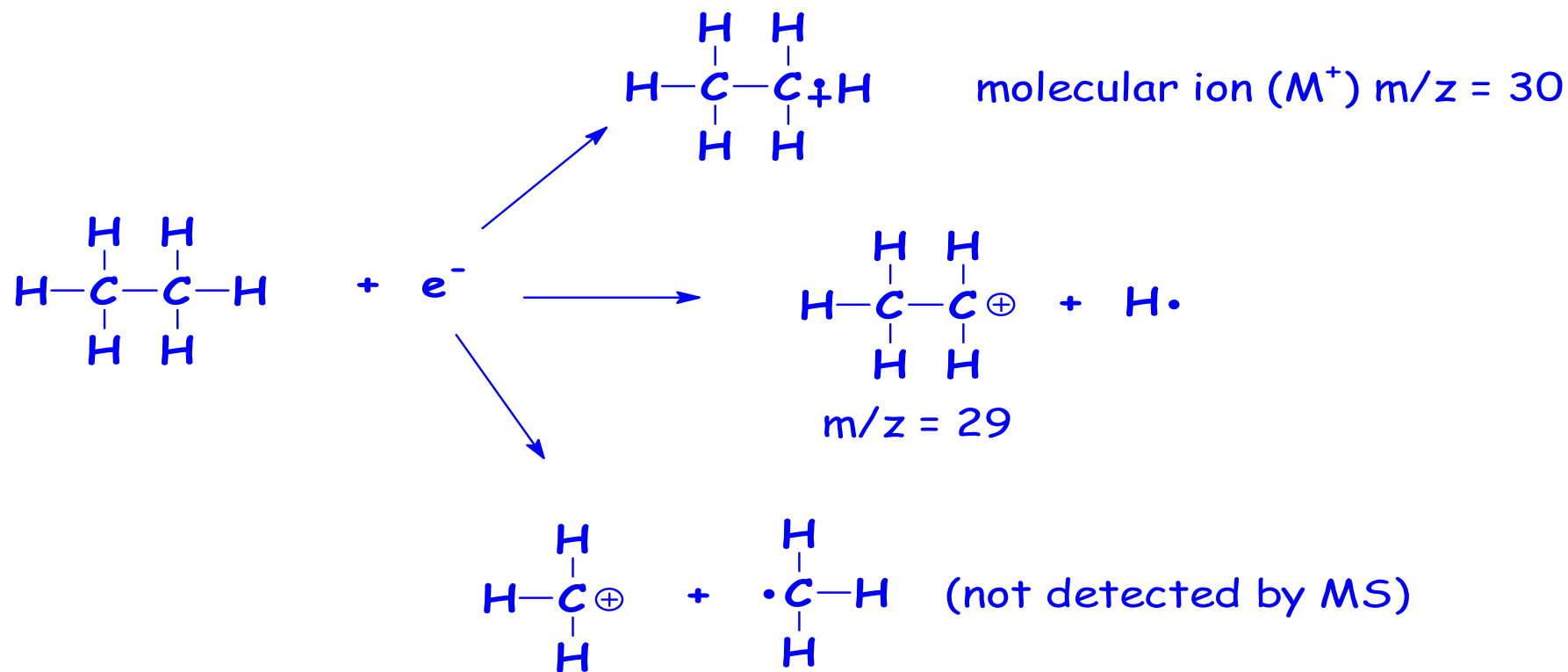
Ionization

- The impact of a stream of high energy electrons causes the molecule to lose an electron forming a **radical cation**.
- A species with a positive charge and one unpaired electron



Fundamental Processes Involved in MS.....

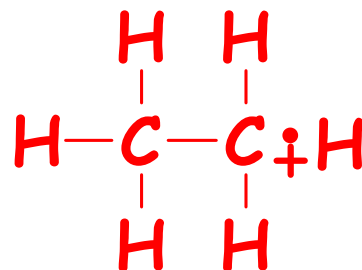
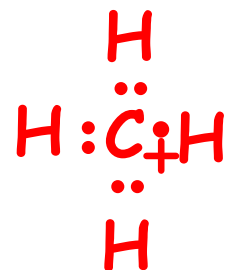
- The formed radical cation is highly **unstable**— it is again broken into fragments



Fundamental Processes Involved in MS.....

- **Molecular ion (parent ion):**

- The radical cation corresponding to the mass of the original molecule



- **The molecular ion is usually the highest mass in the spectrum**
 - Some exceptions with specific isotopes
 - Some molecular ion peaks are absent.

Fundamental Processes Involved in MS.....

3. Separation of the ions is carried out in a curved metal tube through which the ion beam passes

- ✓ *Once formed, the ions are immediately extracted from this part of the instrument to be focussed and accelerated by a series of **electronic lenses**, to increase their kinetic energy*
- ✓ *The ions are then '**filtered**' by the analyser according to their mass-to-charge ratio*

Fundamental Processes Involved in MS.....

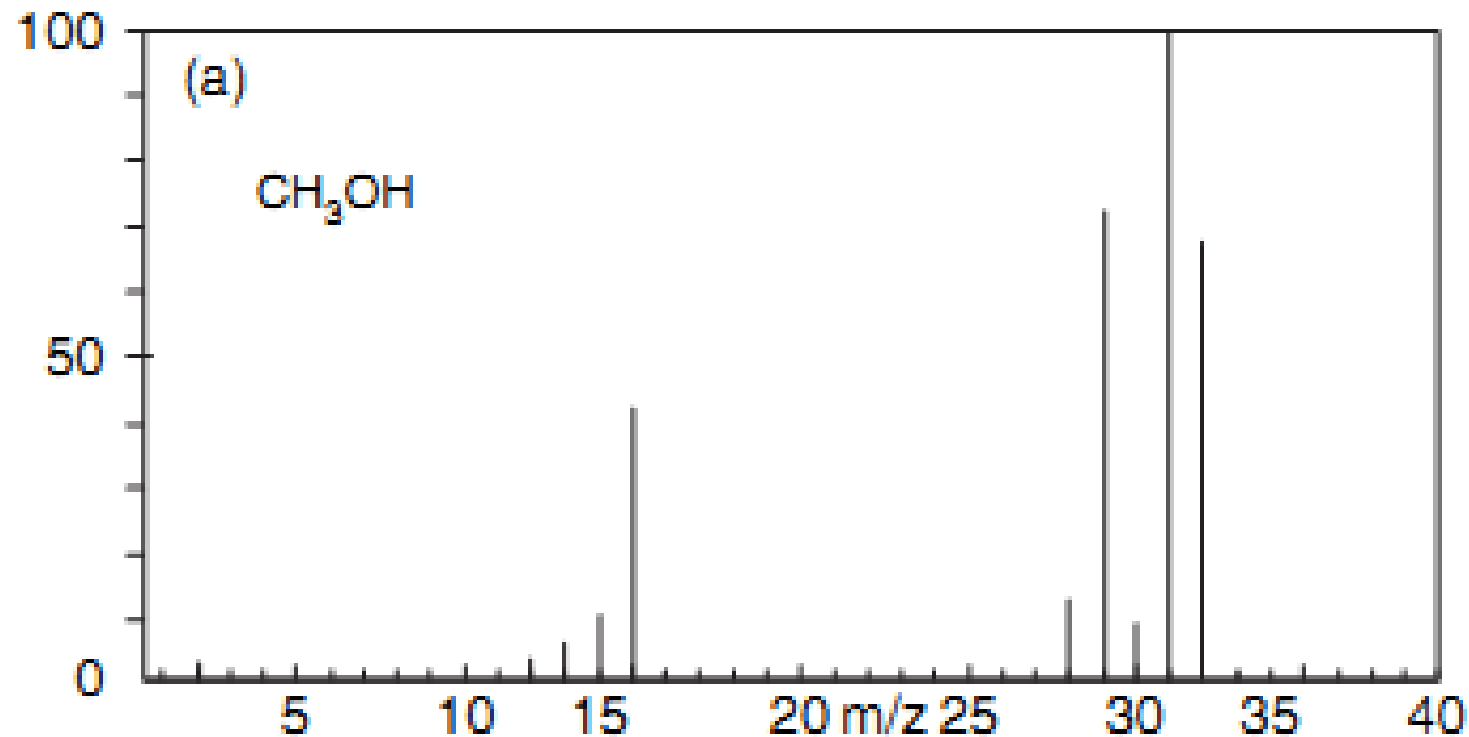
4. Detection

- After separation, ions terminate their flight by striking the sensor of the detector which measures electrical charge and amplifies the weak ionic current.

✚ **NB.** *Mass spectra must be recorded under conditions of high vacuum otherwise collisions between the ions and molecules of atmospheric gases would interfere with the results.*

Fundamental Processes Involved in MS.....

5. Display of the mass spectrum arising from treatment of the signal sent by the detector.



IONIZATION METHODS

Most common ionizers

- **Electron impact (EI),**
 - **Chemical ionization (CI)**
 - **Electrospray (ESI)**
 - **Fast Atom Bombardment (FAB)**
 - **Atmospheric Pressure Chemical Ionization (APCI)**
 - **MALDI (Matrix Assisted Laser desorption ionization)**
- 

Reading Assignment

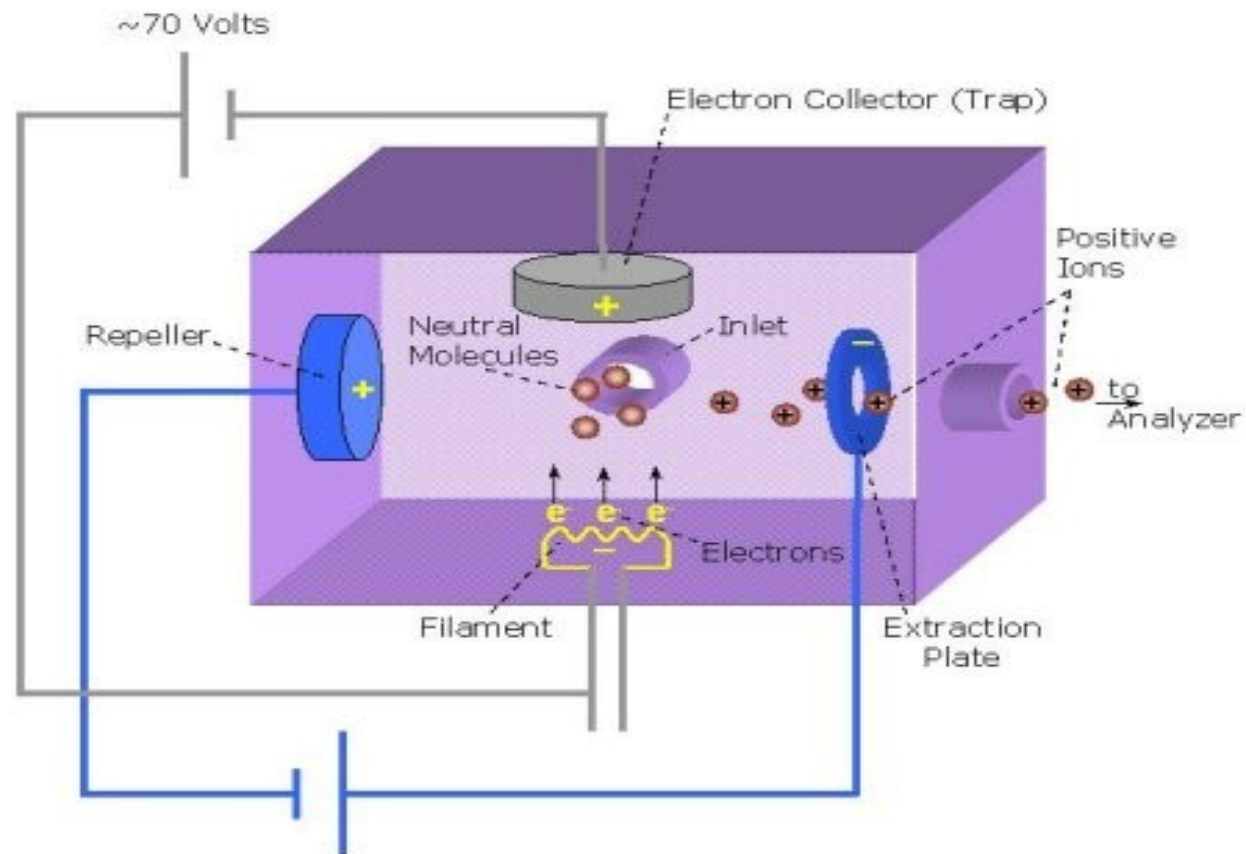
IONIZATION METHODS....

Electron Impact Ionization (EI)

- ✚ Molecules in the **gaseous state under low pressure** are **bombarded** with **a beam of high energy electrons (about 70 electron-volts; eV)**
- ✚ This **bombardment** can first dislodge one of the electrons of the molecule and produce a positively charged ion called **“The molecular ion (M^+)”**

IONIZATION....

Electron Impact Ionization Source

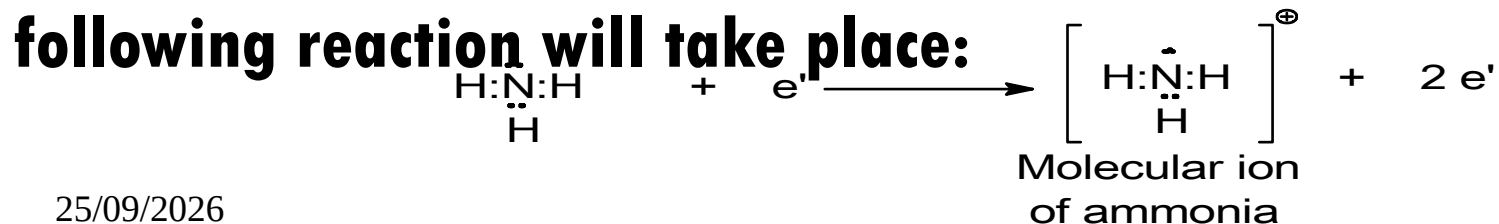


IONIZATION....



$\begin{array}{c} \text{+} \\ \text{+} \\ \text{+} \end{array}$ The molecular ion contains an odd number of electrons, thus it is a **Free radical**, or generally **“Radical Cation”**

$\begin{array}{c} \text{+} \\ \text{+} \\ \text{+} \end{array}$ **Example:** If the molecule under bombardment is a molecule of ammonia, the



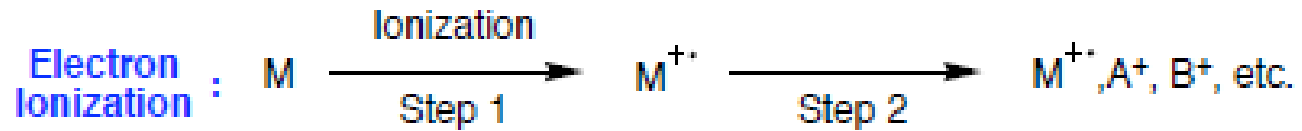
IONIZATION....

Note that:

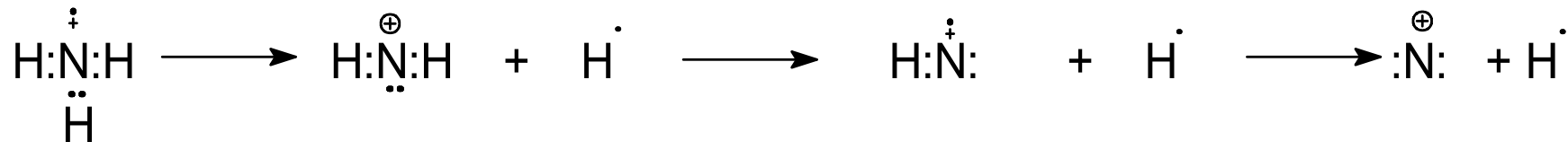
- ✓ **An electron beam with an energy of about 70 eV (about 1600 Kcal./mole) not only produces molecular ions but also imparts to the molecular ions considerable amount of surplus energy.**
- ✓ **This surplus energy is quite sufficient to break any covalent bond in the molecular ion (50 – 100 Kcal/mole)**

IONIZATION....

- Thus soon after they are formed, most molecular ions **undergo fragmentation**
- Fragmentation can take place in a variety of ways depending on the nature of the molecular ion



- It can give us highly useful information about the **structure of a complex molecule**



IONIZATION....

Chemical Impact Ionization

✚ Ionization begins when a reagent gas (R) is ionized by electron impact and then subsequently reacts with analyte molecules (M) to produce analyte ions

✚ e.g. CH_4 , NH_3

CI using CH_4 as reagent

Primary ion formation



Secondary reagent ions



IONIZATION....

Product ion formation



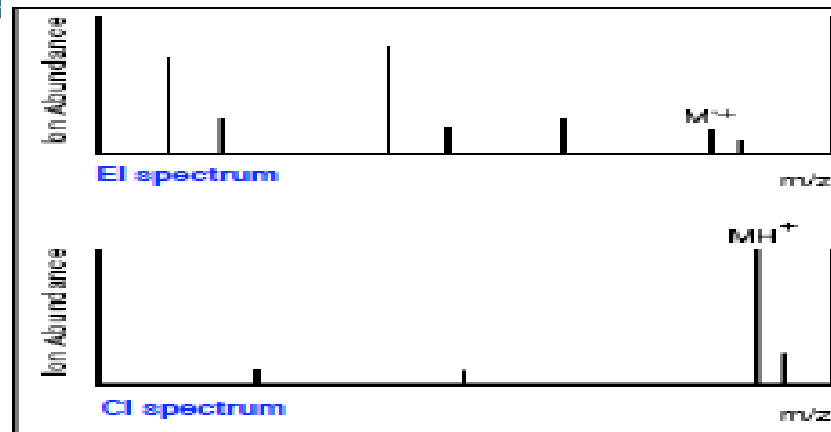
Table 1. Reagent species and their properties for the common PCI gases.

Reagent Gas	Major Species	CI Products	Proton Affinity (kcal/mole)
Methane	CH_5^+	M+1	131.6
	$C_2H_5^+$	M+29	162.6
	$C_3H_5^+$	M+41	—
Isobutane	$C_4H_9^+$	M+1	195.9
Ammonia	NH_4^+	M+1	204.0
		M+18	—

✚ This method gives **molecular weight information** and **reduced fragmentation** in comparison to **EI**.

IONIZATION....

- The two types of spectra that resulted are useful
 - CI sources allow the ready determination of M_r of the analyte and
 - the more complex spectra from EI sources allow for unambiguous identification of the structure of the cpd.



Separation of Ions

- The mass spectrometer sorts the produced cations on the basis of their **masses/charge ratio (m/z)**
- Since the charge on all of the ions = +1, thus the sorting of them actually on the basis of their masses.

Separation of Ions...

- This is done by accelerating the ions through a series of slits, then ions beam passes through a **Curved Tube**
- This curved tube (**Analyzer tube**) passes through a **variable magnetic field**
 - the magnetic field exerts an influence on the moving ions
- *i.e analyzer tube is surrounded by a magnet whose magnetic field deflects the positively charged fragments in a curved path.*

Separation of Ions...

- At a given magnetic field strength, the degree to which the path is curved depends on the mass-to-charge ratio (m/z) of the fragment.
- Path of a fragment with **a smaller m/z** value will **bend more** than that of a heavier fragment
- In this way, the particles with the same m/z values can be separated from all the others.

Separation of Ions...

- If a fragment's path matches the curvature of analyzer tube, the fragment will pass through the tube and out the ion exit slit (**These ions are said to be in register**)
- A collector records the relative number of fragments with a particular m/z passing through the slit
- The **more stable the fragment**, the more likely it will make it to the collector
- Strength of magnetic field is gradually increased, so fragments with progressively larger m/e values are guided through tube and out the exit slit

Separation of Ions...

Note that

- ✚ Actual sorting of ions takes place first in the **gradually increased magnetic field**.
- ✚ Ion-sorting can also be done with “**Electrical Focusing**”,
 - ➡ in this technique the **magnetic field is held constant and the accelerating voltage is varied**.
 - ➡ It gives the same sorting results

Mass Spectrum

- The mass spectrometer records a **mass spectrum**:
 - a graph of relative abundance of each fragment (% Intensity) plotted against its *m/z* value.
- Because the charge (*z*) = +1 for all fragments that reach the collector plate so the *m/z* value express molecular mass (*m*) of fragments.

Mass Spectrum

- + Remember that only **positively charged** species reach the collector.
- + The mass spectrum is usually published either as a **bar graph or in table form.**
- + Peak with the highest **m/z** value in the spectrum is due to the fragment that results when an electron is knocked out of a molecule of the injected sample.
- + In other words, the peak with the **highest m/z** value represents the **molecular ion $(M)^+$** of the studied compound.

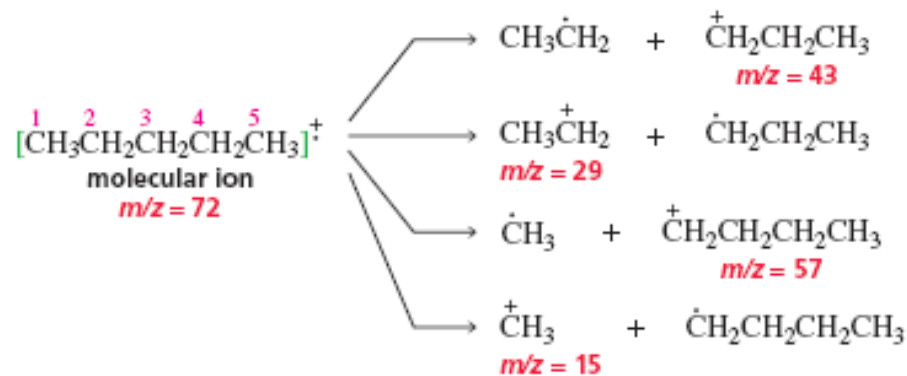
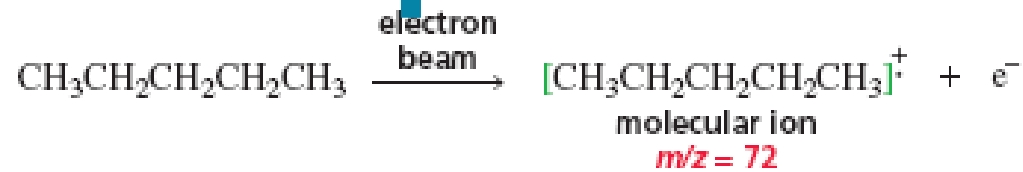
Mass Spectrum cont...

- ✚ Thus m/z value of molecular ion gives the **molecular mass of compound**.
- ✚ Peaks with **smaller m/z values** called **fragment ion peaks** represent **positively charged fragments** of the molecule.

Mass Spectrum cont...

- ✚ “**Base peak**” is the **peak with the greatest intensity**, due to its having the **greatest relative abundance**.
- ✚ The base peak is assigned a **relative intensity of 100%**, and the relative intensity of each of the other peaks is reported as a **percentage of the base peak**.

Mass Spectrum cont...



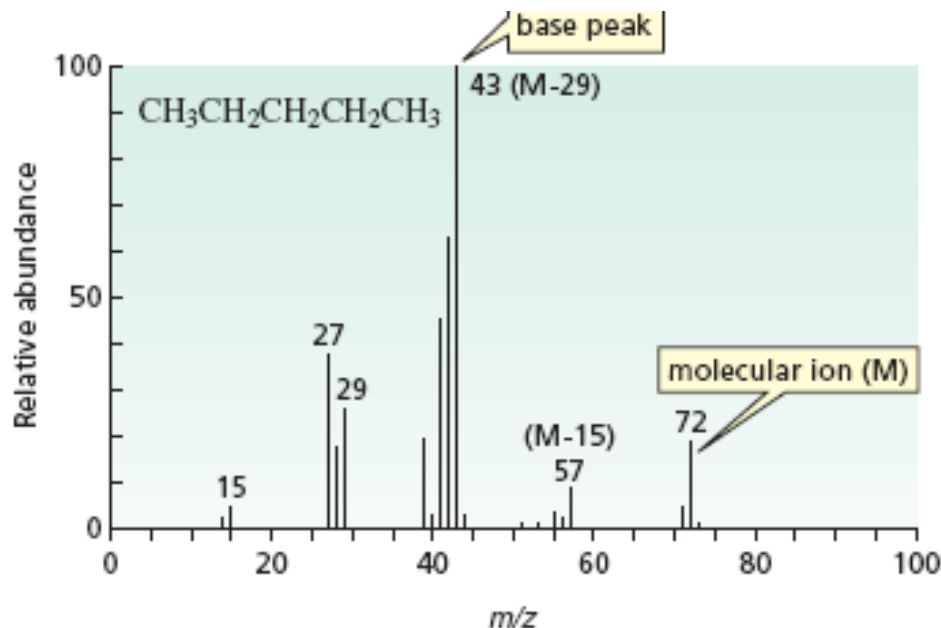
Example: Pentane molecule:

Mass spectrum of pentane, shown as a bar graph and in tabular form.

Base peak represents fragment that appears in greatest abundance. Value of molecular ion gives molecular mass of the compound.

Note that:

The way by which the molecular ion to be fragmented depends on the strength of its bonds and stability of the fragments.



m/z	relative abundance
73	0.52
72	18.56
71	4.32
57	11.20
43	100.00
42	55.27
41	37.93
39	12.44
29	26.65
28	17.75
27	31.22
15	4.22
14	2.56

Mass Spectrum cont...

Note that:

1- In some cases the **base peak** is the **molecular ion peak**

▶ but in most cases it is different from that of the molecular ion

▶ according to the **relative stability of ions** during fragmentation

Mass Spectrum cont...

- 2- Peaks are commonly observed at m/z values **one and two units less** than the m/z values of the carbocations
- ▶ because the carbocations can undergo further fragmentation by losing **one or two hydrogen atoms**

Mass Spectrum cont...

3- Small peak that occurs at m/e $M+1$ and $M+2$ are one and two mass unit greater than the molecular ion(M^{+}),

➡ The $M+1$ and $M+2$ peaks appear because most of elements have more than one **naturally occurring isotope**.

Mass Spectrum cont...

- Principal stable isotops of common elements:

Elements	Isotope	Relative Abundance	Isotope	Relative Abundance	Isotope	Relative Abundance
Carbon	^{12}C	100	^{13}C	1.11		
Hydrogen	^1H	100	^2H	0.016		
Nitrogen	^{14}N	100	^{15}N	0.38		
Oxygen	^{16}O	100	^{17}O	0.04	^{18}O	0.20
Fluorine	^{19}F	100				
Silicon	^{28}Si	100	^{29}Si	5.10	^{30}Si	3.35
Phosphorus	^{31}P	100				
Sulfur	^{32}S	100	^{33}S	0.78	^{34}S	4.40
Chlorine	^{35}Cl	100			^{37}Cl	32.5
Bromine	^{79}Br	100			^{81}Br	98.0
Iodine	^{127}I	100				

Mass Spectrum cont...

✚ From the table above

➡ For C, H, N and the principal heavier isotope is **M+1**

➡ For O, Si, S, Cl and Br , the principal heavier isotope is the **M+2**

➡ For F and I, there is no heavier isotopes (not affect M+1 or M+2)

Guides to determine molecular formula : for organic compounds:

Nitrogen rule:

- ✚ If molecular ion (M^+) is even number, compound contains an even number of N atoms (or zero nitrogen atom)
- ✚ If the M^+ peak is odd, the molecule contains an odd number of nitrogens
- ✚ All other common atoms in an organic compound give a mass which is **even**.

The index of hydrogen deficiency

- ➡ Due to double and triple bonds and cyclization
- ➡ can be calculated for the compounds containing C, H, N, O, S & halogens from the formula:
- ➡ $\text{Index} = \text{Carbons} + 1 - \text{Hydrogens}/2 - \text{Halogens}/2 + \text{Nitrogens}/2$
- ➡ Divalent atoms like O and S are not counted in formula

Fragmentation

- ✚ Although the **molecular ion** is useful for identification, it does not provide any structural information about an unknown cpd.
- ✚ The structural information is obtained from the **fragmentation patterns** of the mass spectrum.

Fragmentation cont...

- ✚ After a molecule is ionized, the molecular ion retains the **excess ionization energy**.
- ✚ If this excess energy is greater than the energy required to break a chemical bond, the molecule can fragment.
- ✚ The fragmentation processes are typically categorized as direct **cleavage** or **rearrangement**.

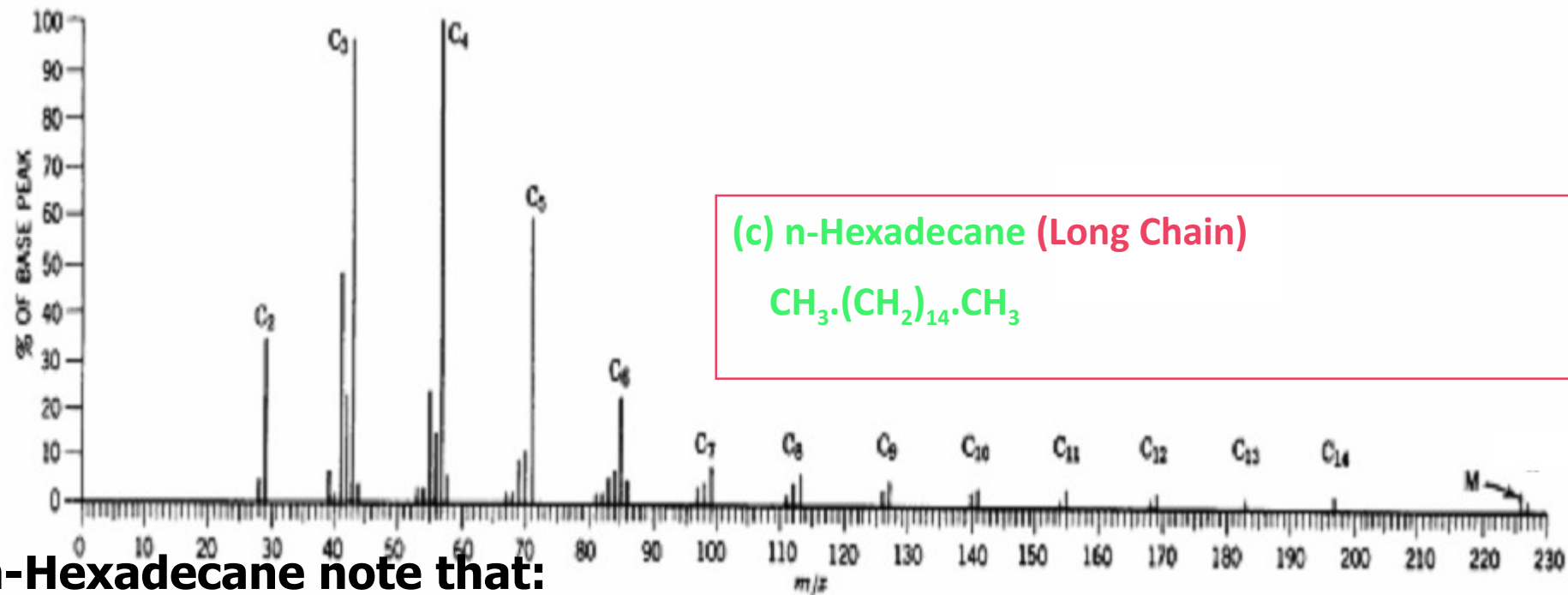
Fragmentation rules in MS

1. Non Branched saturated Hydrocarbons:

- ✚ Intensity of M^+ is Larger for **linear** chain than for **branched** compound
- ✚ Intensity of M^+ decrease with Increasing M.W.
- ✚ Molecular ion peak of a **straight-chain saturated** hydrocarbon is always **present**
 - But of low intensity for the long chain compounds

Non Branched saturated Hydrocarbons...

- ✚ Fragmentation is characterized by presence of clusters of peaks and
 - the corresponding peaks of each cluster are 14 (CH_2) mass units apart
- ✚ The most abundant (Stable) fragments are at **C₃** and **C₄**
- ✚ Simple cleavage at a single bond , can take place in at least two ways
 - each way produces a **cation and a free radical**



For n-Hexadecane note that:

- A very small molecular peak is present (C₁₆).

- The C₁₅ fragment is not present or of very small abundance.

- Fragments from C₁₄ down to C₇ have small abundances also but with increased values and clustering starts to appear.

- Fragments from C₆ down to C₂ have the most intense peaks (more stable), and C₄ is the base peak (the most stable one)

2. Branched Saturated Hydrocarbons:-

- ✚ Cleavage is favored at branching

 - ➡ reflecting the Increased stability of the ion

- ✚ the more substituted , the more likely is cleavage

 - ➡ This is a consequence of the increased stability of a tertiary carbocation over a secondary , which in turn is more stable than a primary one.

- ✚ Generally, the **largest substituent** at a branch is eliminated most readily as a **radical**

 - presumably because a long-chain radical can achieve some stability by delocalization of the lone electron

Non Branched saturated Hydrocarbons...

Examples: (a) 2-Methylbutane:-



Secondary carbocation, ($M^+ - 15$)

Secondary carbocation, ($M^+ - 29$)

Abundant ($m/e = 57$)

Most stable (base peak), $m/e = 43$



Molecular ion, small peak, $m/e = 72$



Abundant (Primary carbocation)

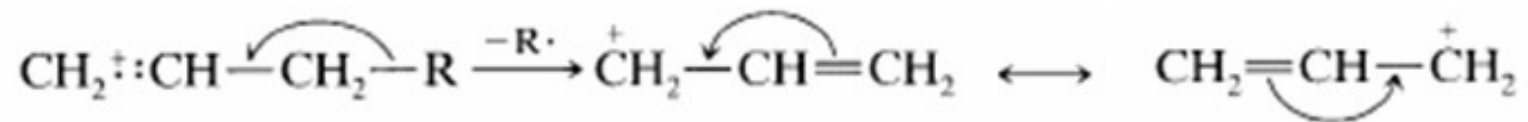
Small (Primary carbocation)

$m/e = 29 (M^+ - 43)$

$m/e = 15 (M^+ - 57)$

Fragmentation rules in MS cont...

3. Double bonds, cyclic structures especially aromatic or hetero-aromatic rings stabilize the **molecular ion** and thus increase the probability of its appearance, [e.g.](#) (next slide)
- 4- Double bonds favor **allylic** cleavage and give the resonance-stabilized allylic carbocation



Fragmentation rules in MS cont...

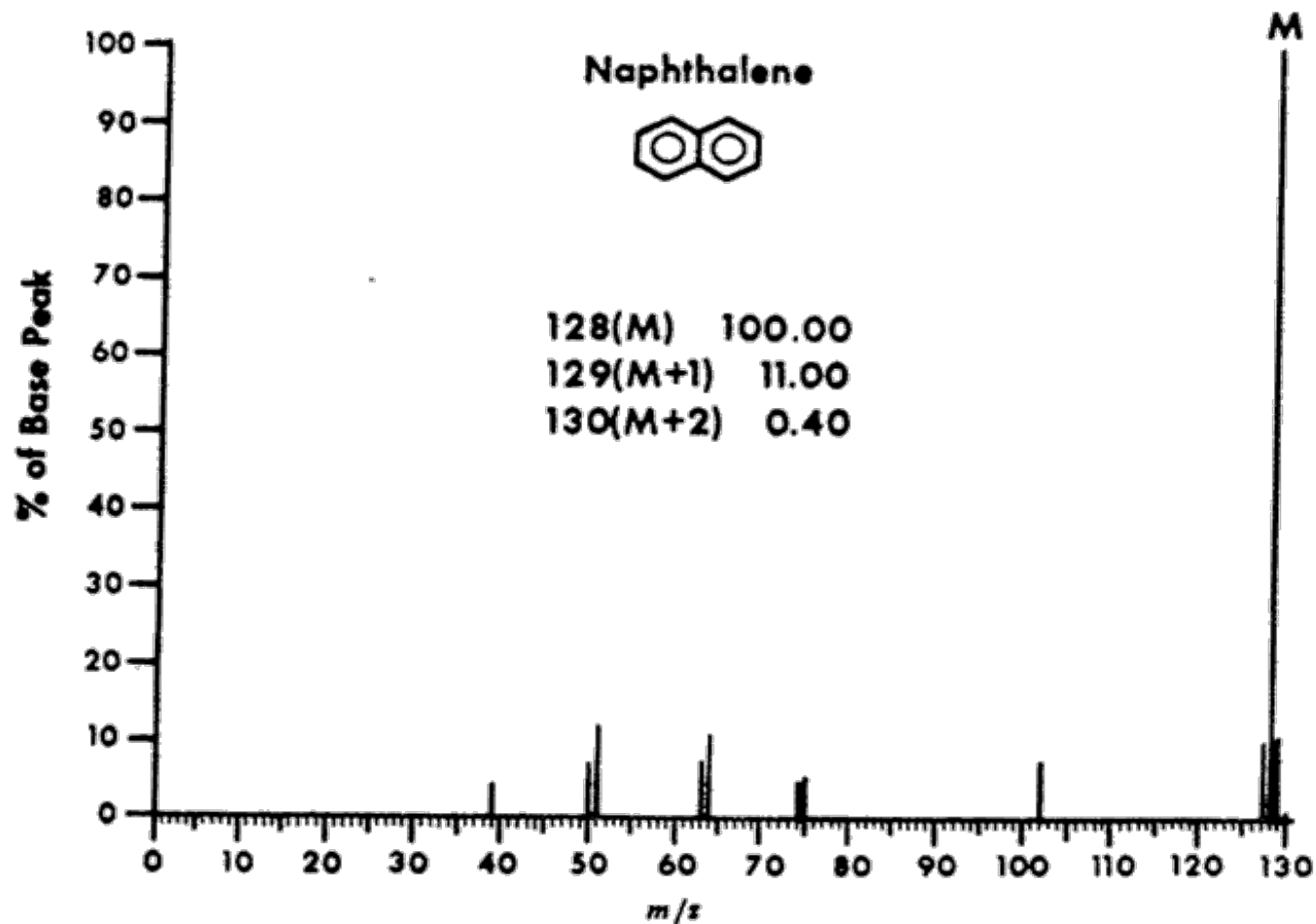


FIGURE 2.8. Naphthalene.

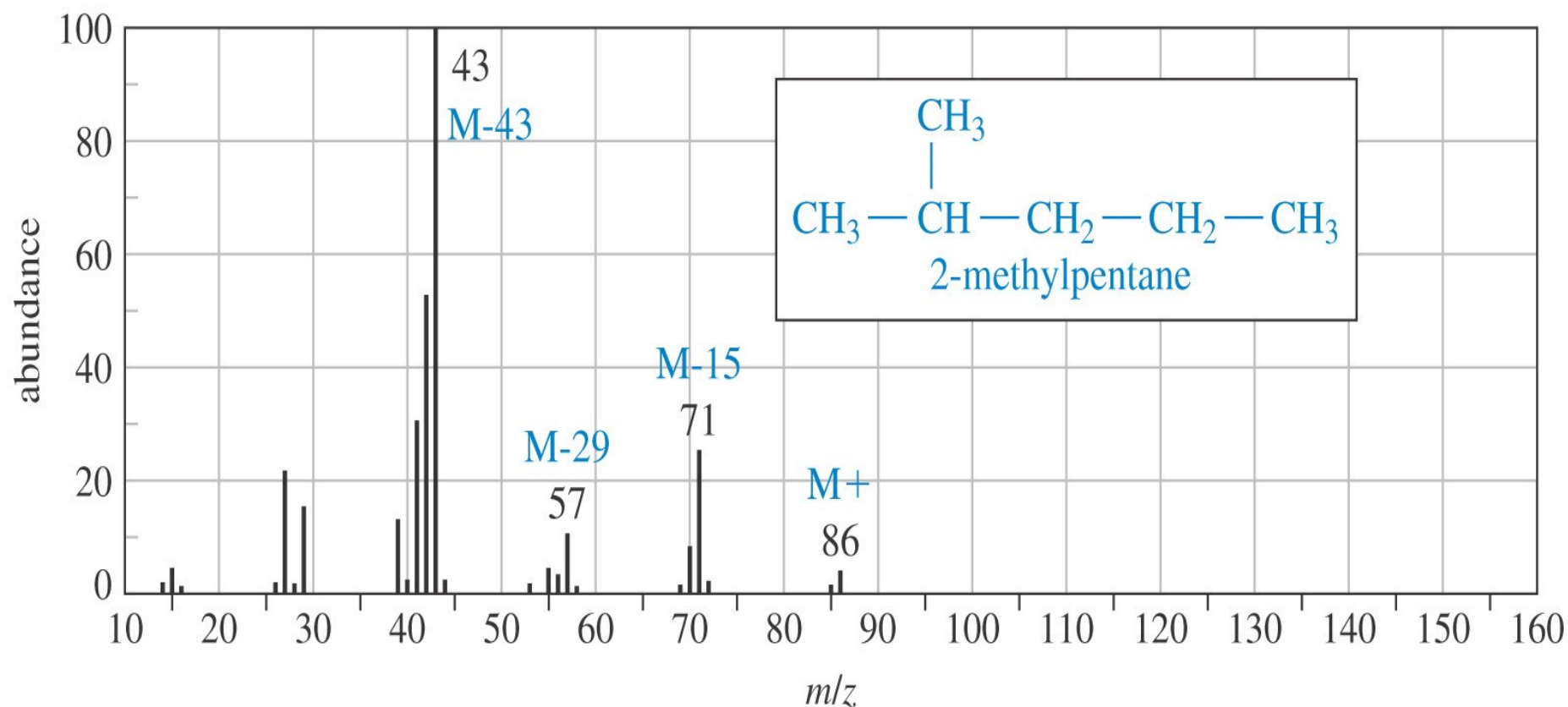
Fragmentation Patterns...

Alkanes

- Fragmentation often splits off simple alkyl groups:
 - Loss of methyl $M^+ - 15$
 - Loss of ethyl $M^+ - 29$
 - Loss of propyl $M^+ - 43$
 - Loss of butyl $M^+ - 57$
- Branched alkanes tend to fragment forming the most stable carbocations.

Fragmentation Patterns...

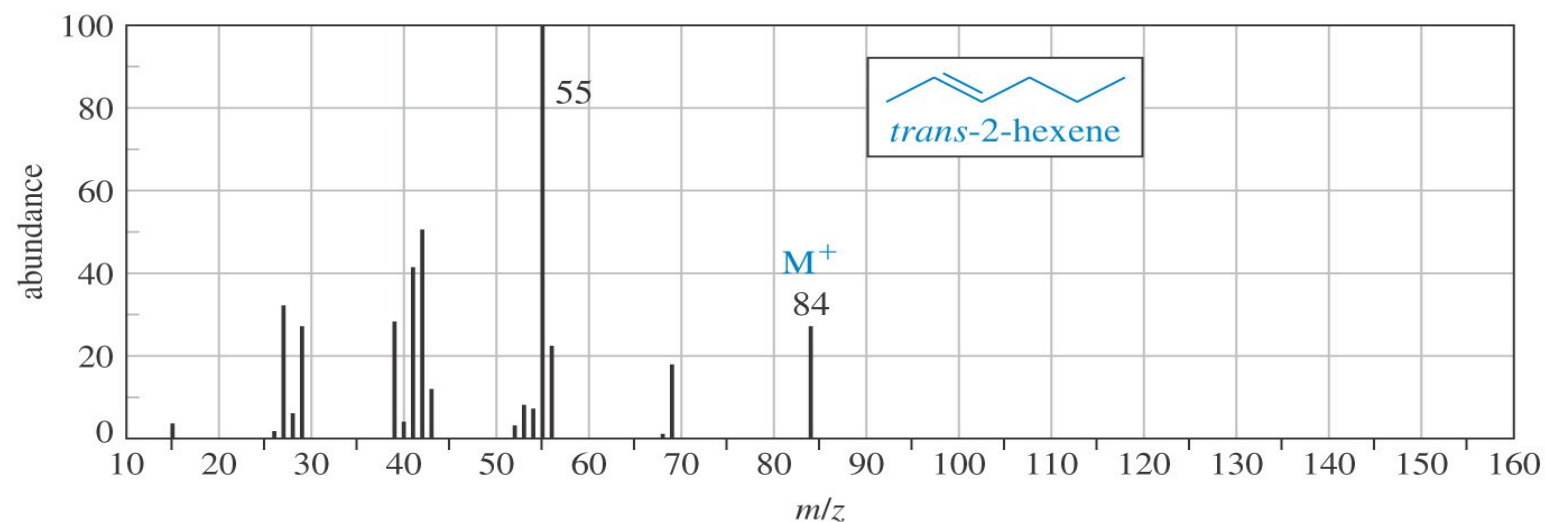
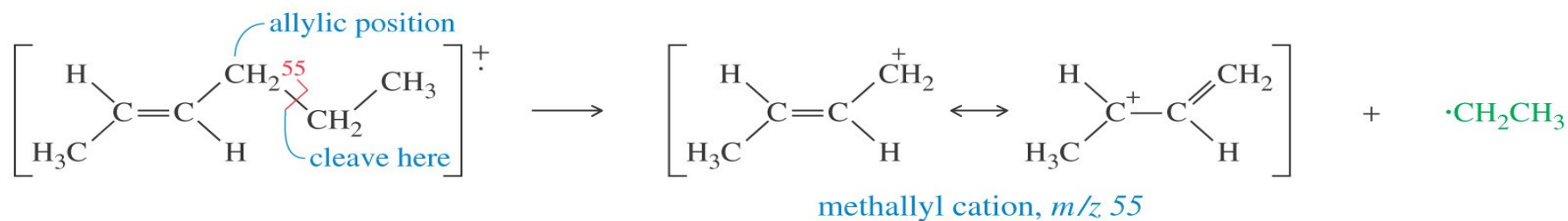
Mass spectrum of 2-methylpentane



Fragmentation Patterns...

Alkenes:

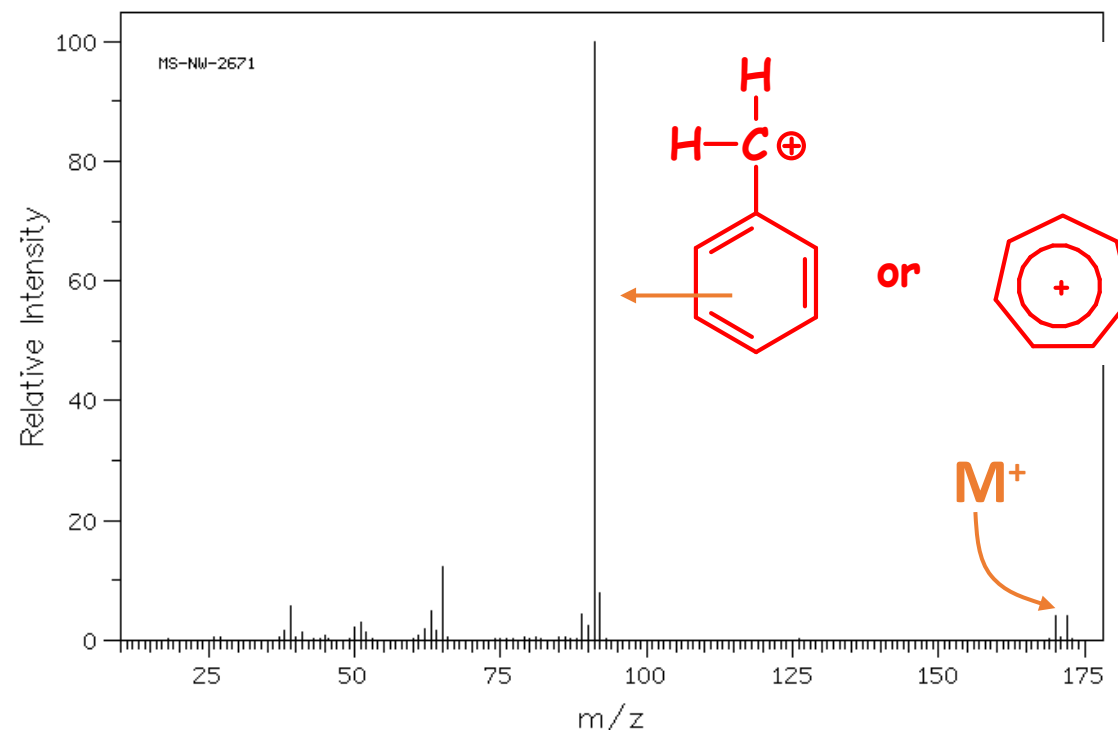
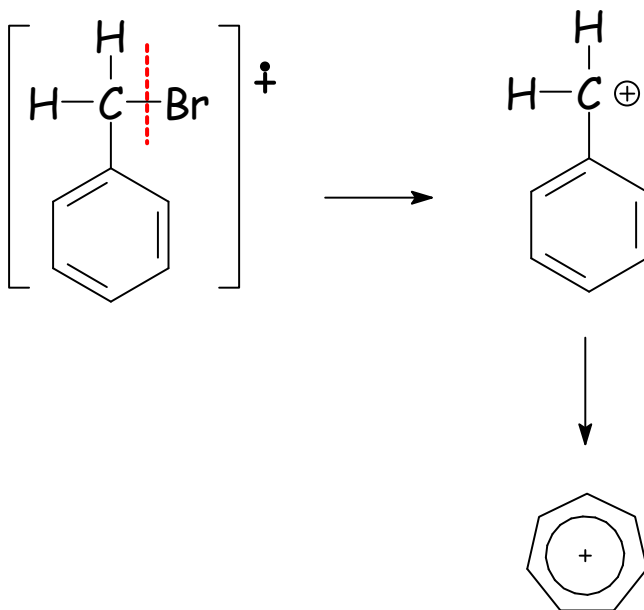
- Fragmentation typically forms resonance stabilized allylic carbocations



Fragmentation Patterns...

Aromatics:

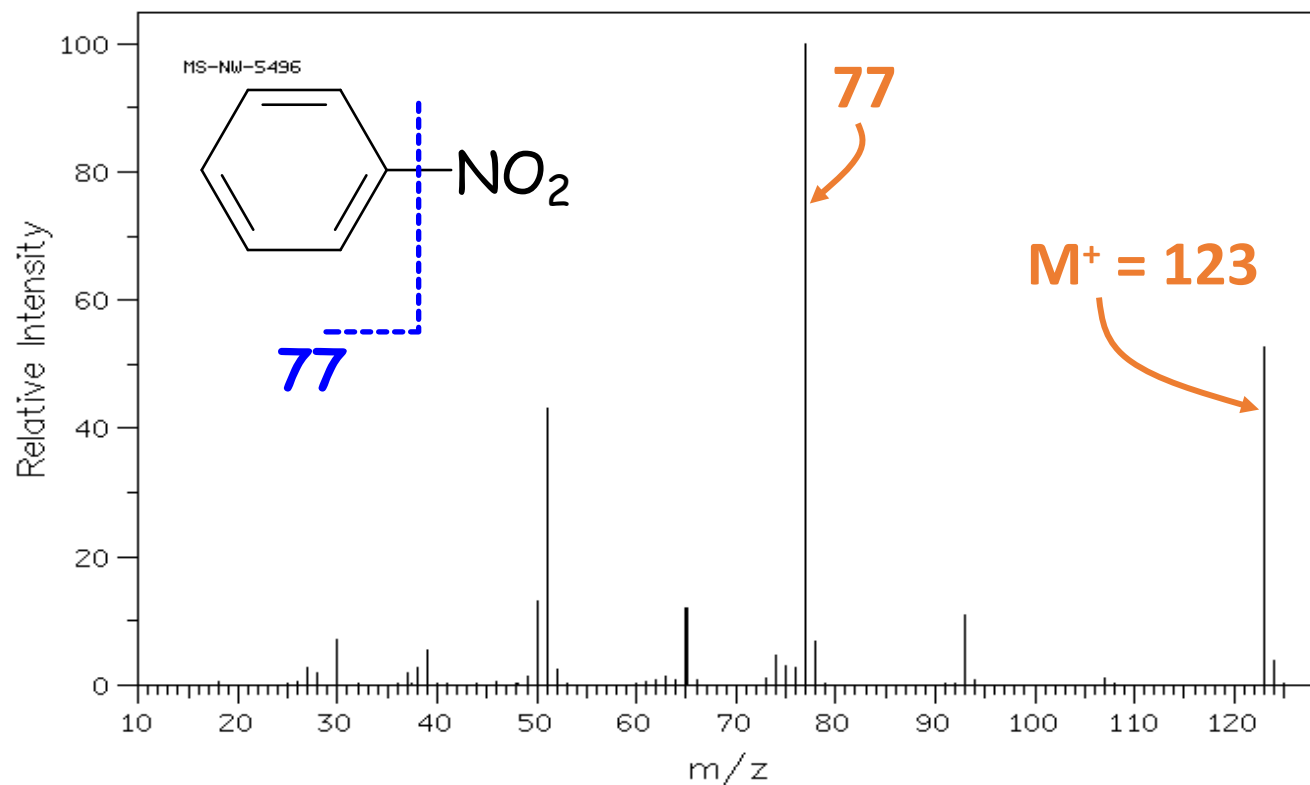
- Fragment at the **benzylic carbon**, forming a resonance stabilized benzylic carbocation (which rearranges to the **tropylium ion**)



Fragmentation Patterns...

Aromatics...

- Aromatics may also have a peak at $m/z = 77$ for the benzene ring.



Fragmentation Patterns...

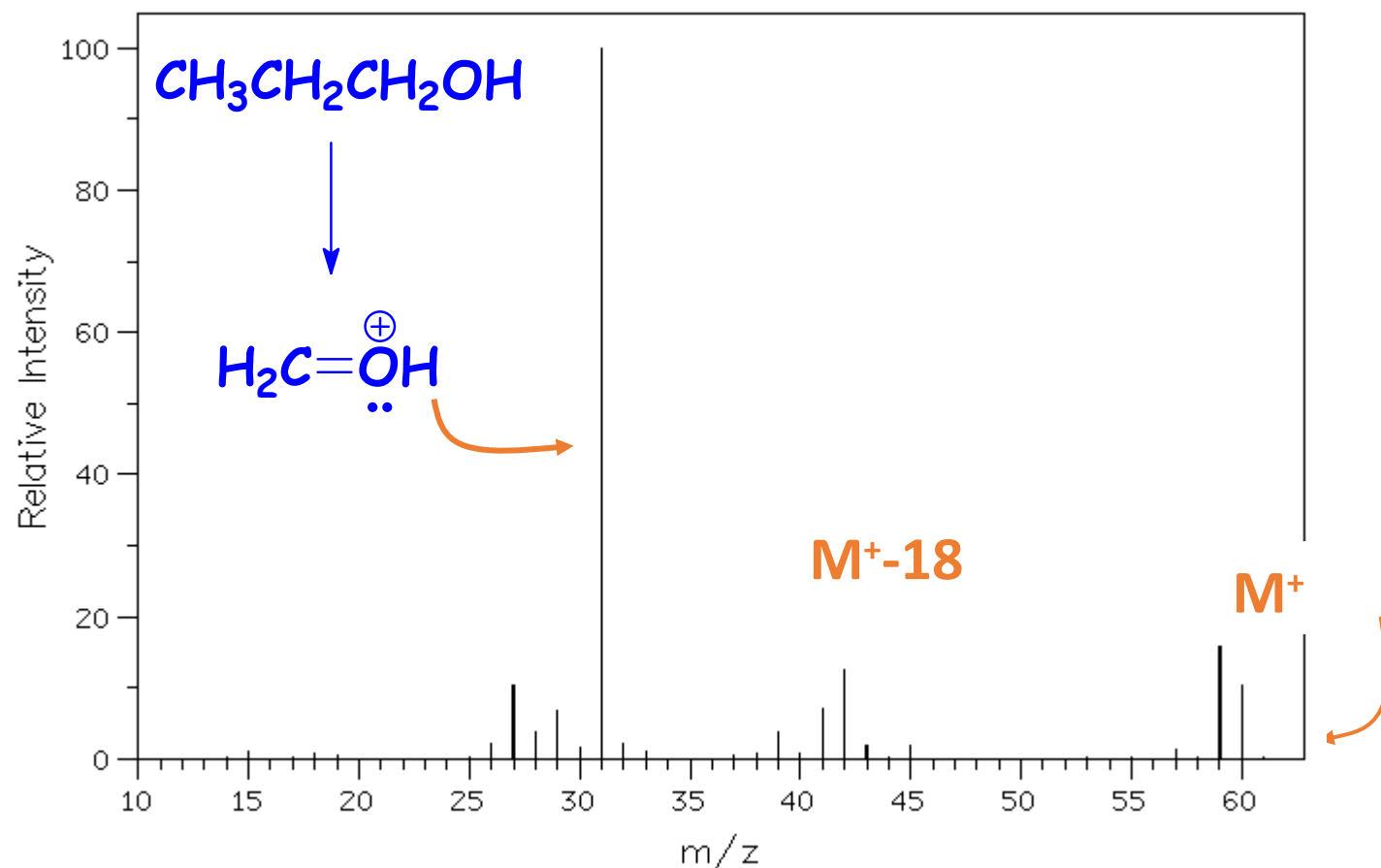
Alcohols...

- Fragment easily resulting in very small or missing parent ion peak
- May lose hydroxyl radical or water
 - $M^+ - 17$ or $M^+ - 18$
- Commonly lose an alkyl group attached to the **carbinol carbon** forming an **oxonium ion**.
 - 1° alcohol usually has prominent peak at $m/z = 31$ corresponding to $H_2C=OH^+$

Fragmentation Patterns...

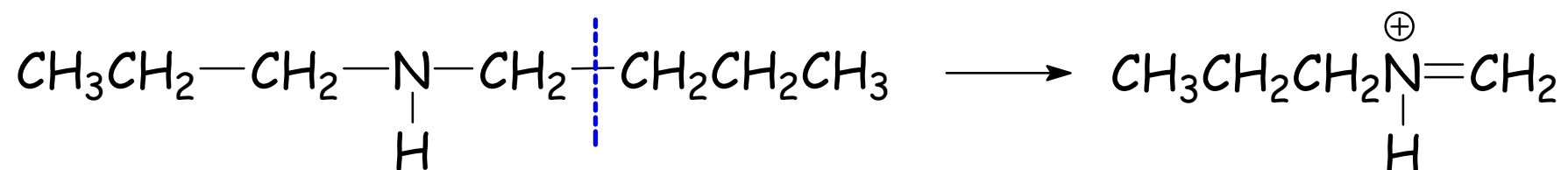
Alcohols...

- **MS for 1-propanol**



Fragmentation Patterns...

- **Amines**
- **Odd M^+** (assuming an odd number of nitrogens are present)
- **α -cleavage dominates forming an iminium ion**

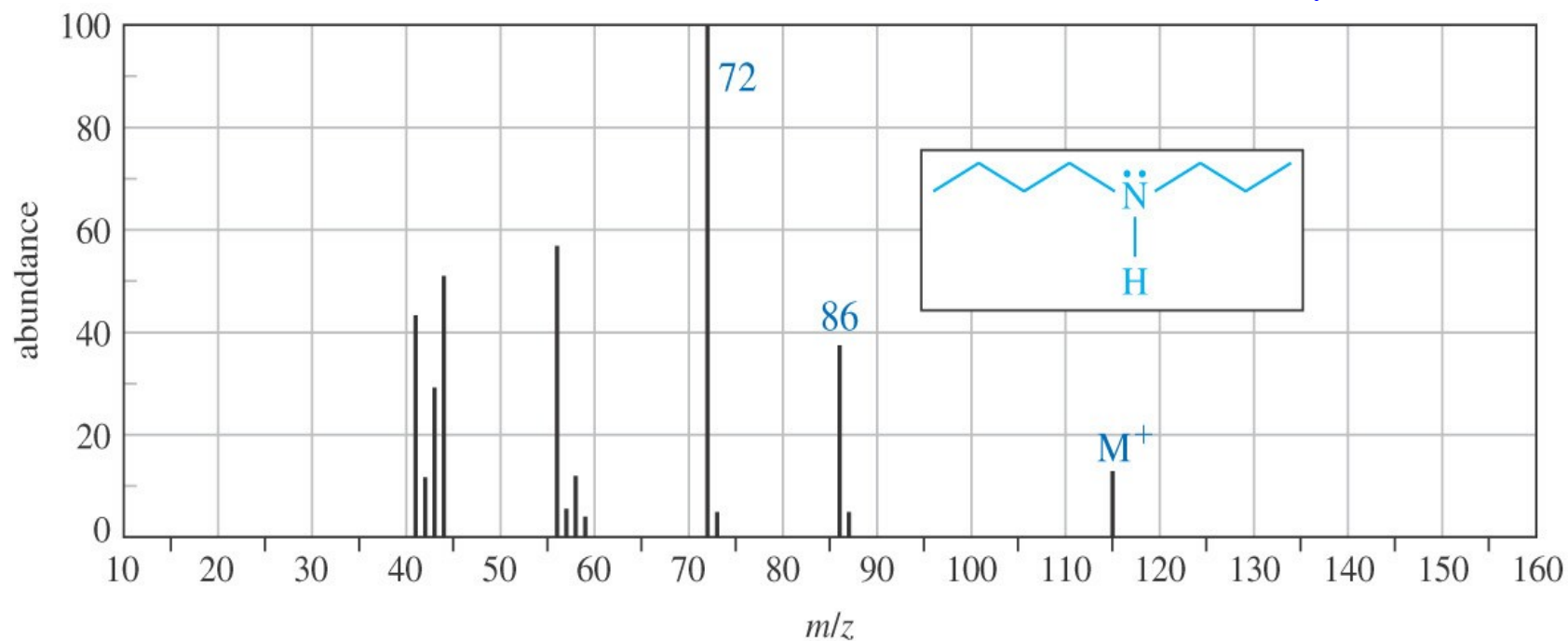
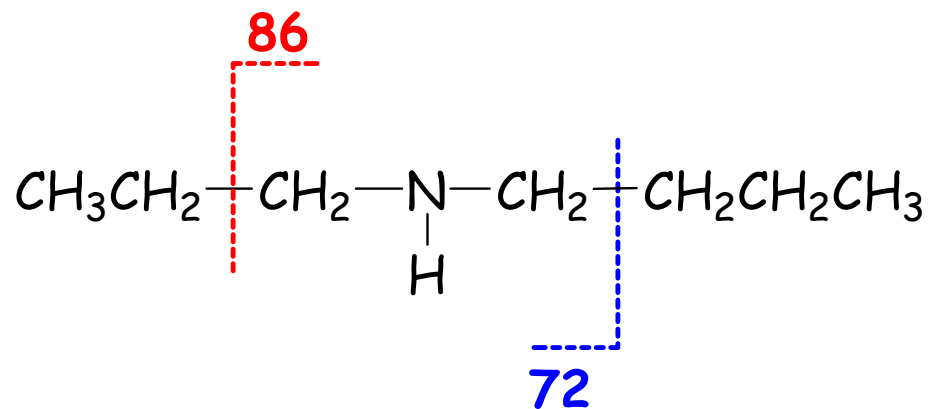


$m/z = 72$

iminium ion

Fragmentation Patterns...

Amines...



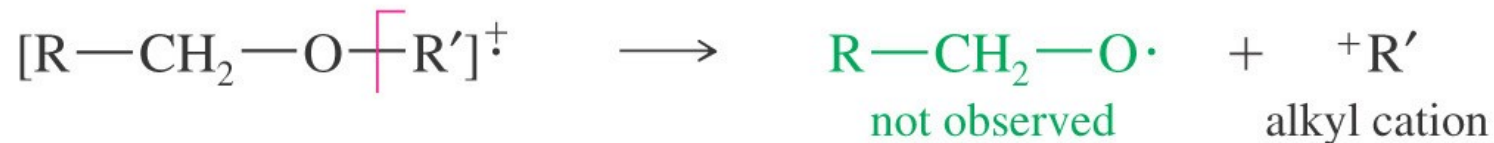
Fragmentation Patterns...

- Ethers

- α -cleavage forming oxonium ion
- Loss of alkyl group forming oxonium ion
- Loss of alkyl group forming a carbocation



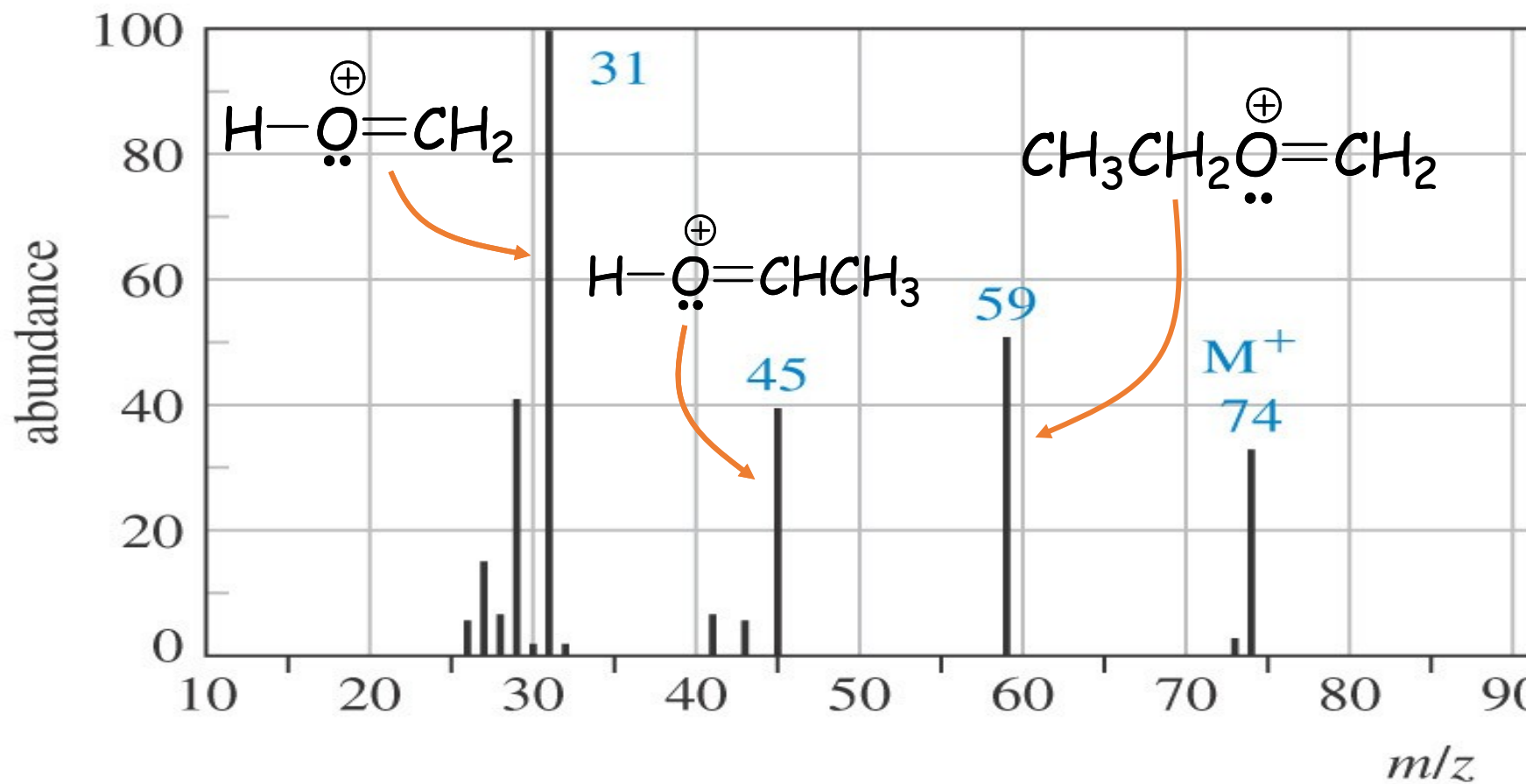
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Fragmentation Patterns...

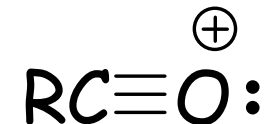
MS of diethylether ($\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$)



Fragmentation Patterns...

Aldehydes (RCHO)

- Fragmentation may form acylium ion

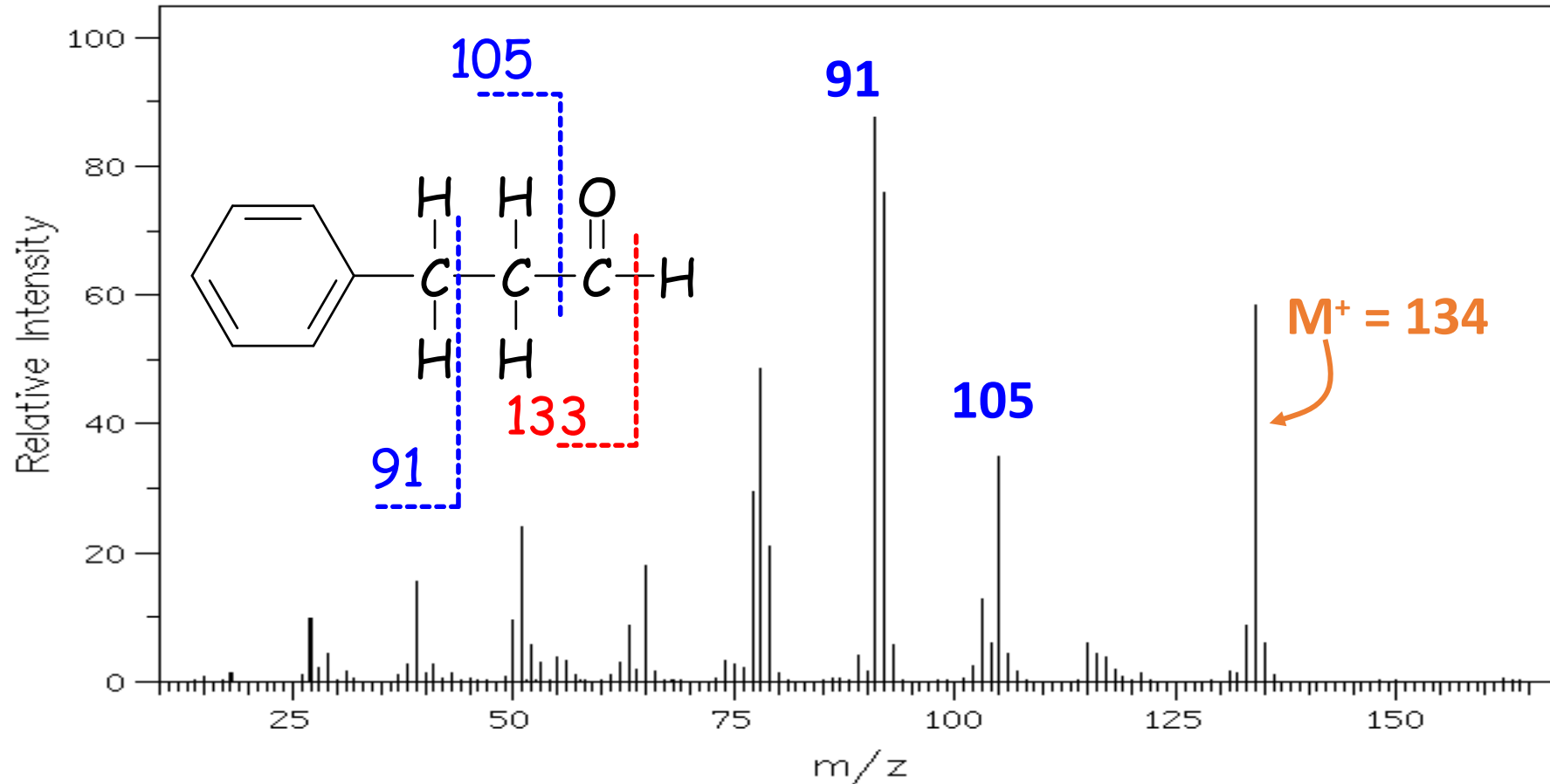


- Common fragments:



Fragmentation Patterns...

- MS for hydrocinnamaldehyde



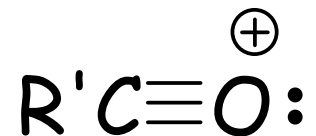
Fragmentation Patterns...

- Ketones

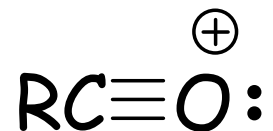


- Fragmentation leads to formation of acylium ion:

- Loss of R forming

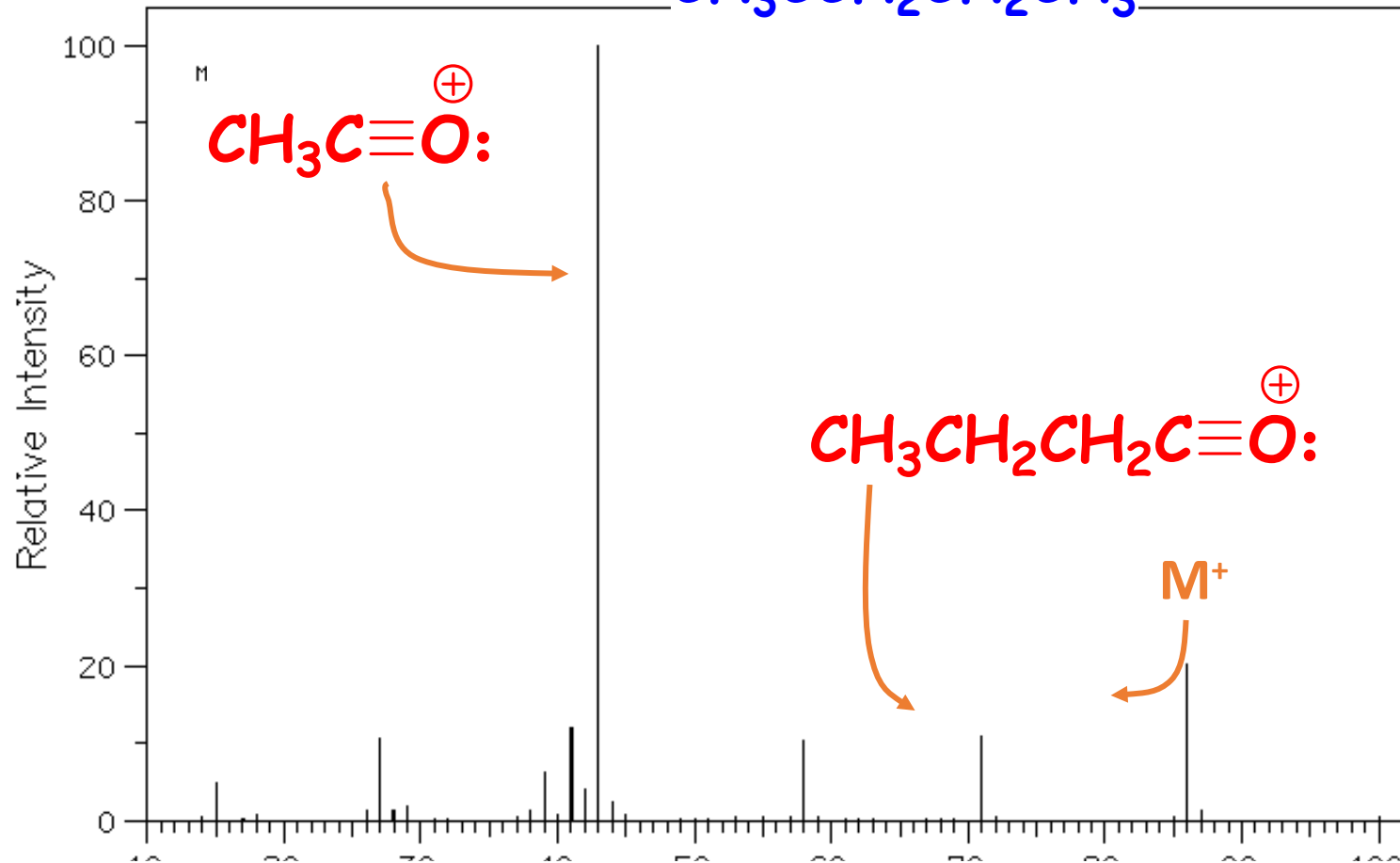


- Loss of R' forming



Fragmentation Patterns...

- MS for 2-pentanone



SDBSWeb : <http://riodb01.ibase.aist.go.jp/sdbs/> (National Institute of Advanced Industrial Science and Technology, 11/28/09)

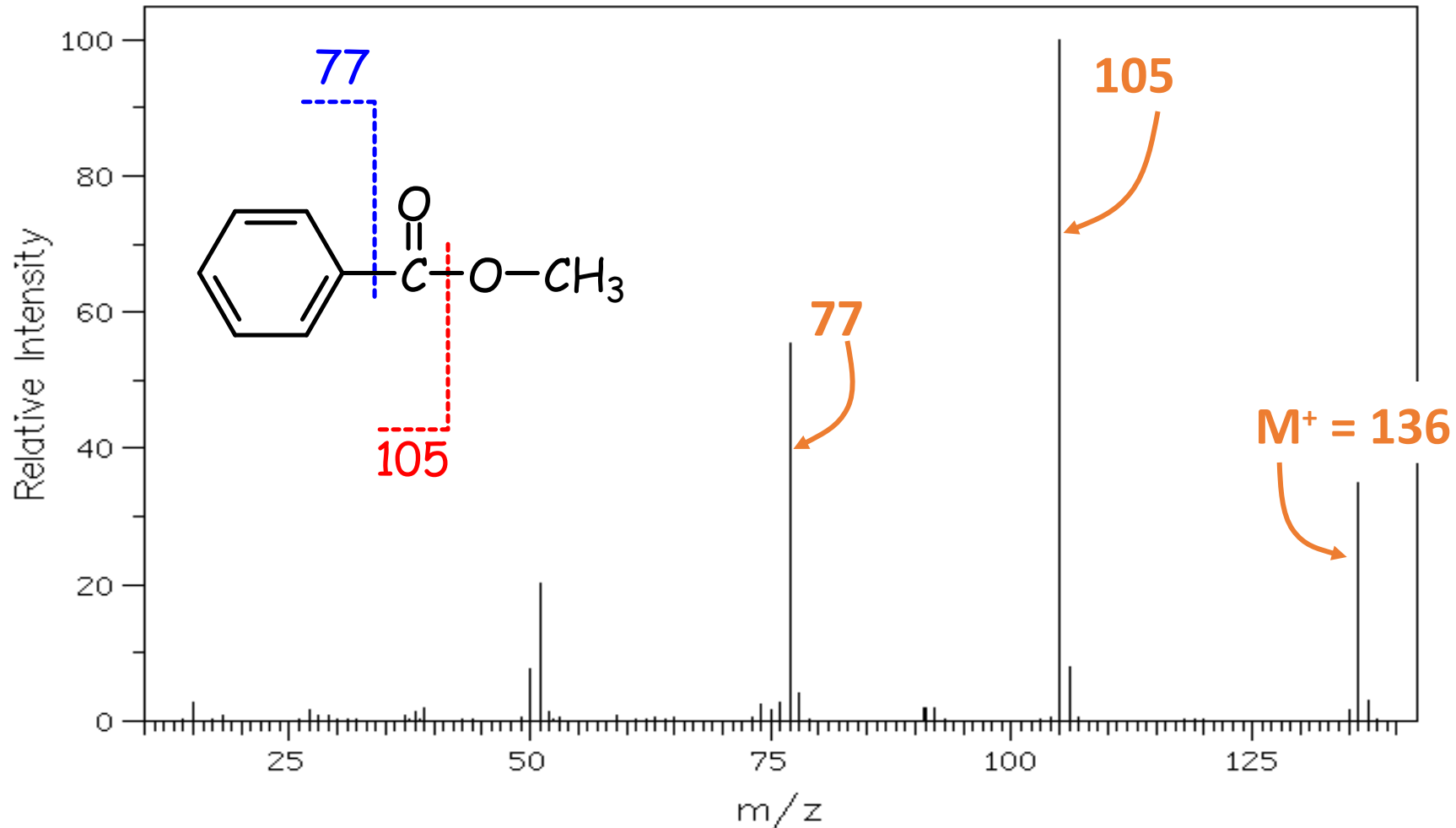
Fragmentation Patterns...

- Esters ($\text{RCO}_2\text{R}'$)
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Rule of Thirteen

- The “Rule of Thirteen” can be used to identify possible molecular formulas for an unknown hydrocarbon, C_nH_m .
 - **Step 1:** $n = M^+/13$ (integer only, use remainder in step 2)
 - **Step 2:** $m = n + \text{remainder from step 1}$

Rule of Thirteen...

- **Example:** The formula for a hydrocarbon with $M^+ = 106$ can be found:

- **Step 1:** $n = 106/13 = 8$ ($R = 2$)

- **Step 2:** $m = 8 + 2 = 10$



Rule of Thirteen...

- If a heteroatom is present,
 - Subtract the mass of each heteroatom from the MW
 - Calculate the formula for the corresponding hydrocarbon
 - Add the heteroatoms to the formula

Rule of Thirteen...

Example: A compound with a molecular ion peak at $m/z = 102$ has a strong peak at 1739 cm^{-1} in its IR spectrum. Determine its molecular formula.

solution: Strong IR absorption at 1739 cm^{-1} Suggests a carbonyl ($\text{C}=\text{O}$) stretch, most consistent with an ester (typically $\sim 1735\text{--}1750\text{ cm}^{-1}$).

you have ester ($-\text{COO}-$) **mass = $12 + 32 = 44$**

□ subtract mass of COO from molecular ion peak mass
ie, 102

$$= 102 - 44 = 58$$

• apply rule of thirteen

- $\text{no C} = 58 \div 13 = 4 = n, r = 6$

- $\text{No H} = n + r = 4 + 6 = 10$

• C_4H_{10} the formula should be adjusted, or add COO

• then chemical formula will be $\text{C}_5\text{H}_{10}\text{O}_2$

Applications in mass spectrometry

1. Identification by means of a spectral library
2. Analysis of the elementary composition of ions (molecular formula determination)
3. Determination of molecular masses from multicharged ions
4. Determination of isotope ratios for an element
5. Identification of organic ions by looking at fragmentation patterns
6. Integration with GC and HPLC (GC-MS, HPLC-MS)

THANK YOU